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Source: *Music Perception: An Interdisciplinary Journal*, Vol. 23, No. 2 (December 2005), pp. 105-118

Published by: University of California Press

Stable URL: https://www.jstor.org/stable/10.1525/mp.2005.23.2.105

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Sensory Consonance: An ERP Study

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the aim of this series of experiments was to determine whether consonant and dissonant chords elicit similar or different electrophysiological effects out of a musical context and whether these effects are similar or different for musicians and nonmusicians. To this end, we recorded the event-related brain potentials (ERPs) elicited by the different intervals of the chromatic scale that were classified into three categories: perfect consonances, imperfect consonances, and dissonances. Participants were to decide, on a six-point scale, whether the intervals evoked pleasant or unpleasant feelings. To test the hypothesis that the perception of dissonance results from the superposition of the partials of close frequencies (Helmholtz, 1877), two notes were either played together (harmonic intervals) or successively (melodic intervals). Since, in this latter case, the two notes are played at different points in time, the perception of roughness, if any, should be weaker than for harmonic intervals. In line with Helmholtz's hypothesis, results showed larger differences for harmonic than for melodic intervals, which were mainly found on the N1-P2 complex for musicians, on the N2 component for nonmusicians, and on a later negative component for both musicians and nonmusicians. However, these results also point to the influence of expertise and cultural factors, since different results were obtained when ERPs were averaged as a function of music theory and according to the participants' responses.

Received January 22, 2002, accepted February 21, 2004

When two notes are played simultaneously, they are perceived differently and they may evoke different sensations depending on the musical interval between them. They can be perceived as consonant and possibly evoke a pleasant sensation, or they can be perceived as dissonant and possibly evoke an unpleasant sensation. Of course, these pleasant/unpleasant sensations may depend on the listener's expertise and familiarity with the type of music and may therefore be highly subjective. Interestingly, however, several theories and hypotheses have been proposed to explain the origin of consonance and dissonance. In the 5th century B.C., Pythagoras was one of the first to notice that two sounds bound by a simple frequency ratio (1:1, 2:1, 3:2, 4:3) produced a more consonant ensemble than more complex ratios (e.g., 16:15). Later, in his well-known theory of consonance, which still remains a reference nowadays, Helmholtz (1877) associated the feeling of dissonance with the presence of beats between pure tones (or partials of complex tones) of neighboring frequencies, giving rise to a sensation of roughness, proportional to the frequency of the beats. Thus, according to Helmholtz, this sensation would have a physiological origin, while still being influenced by other factors, such as subjective and cultural differences. This conception is in line with recent theories that distinguish between sensory consonance and contextual consonance (see Tenney, 1995).

To our knowledge, most of the research on consonance in psychomusicology has been aimed at studying consonance in context, and less out of a musical context (Butler & Daston, 1963; Plomp & Levelt, 1965). However, as Costa, Bitti, and Bonfiglioli (2000) point out, the study of intervals in isolation is important for

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two reasons. First, it controls for some of the elements that influence the perception of consonance (modality, tempo, intensity, rhythm, timbre, etc.). Second, results of certain studies (Balzano & Liesch, 1982; Cohen, Granot, Pratt, & Barneah, 1993) point to the fact that, to a certain extent, the perception of intervals can be independent of the particular context. This conclusion has, however, been challenged by other results. For example, when participants are asked to categorize, on a semantic scale, the 12 possible bichords of a tempered scale, Bozzi (1985) found a clear correspondence between the theoretical description and the results for the dissonant bichords and the octave but not for the consonant bichords. Maher (1980) also showed that, out of the 14 musical intervals used in his experiment, 7 gave no difference for the measure used in the experiment. Thus, results did not clearly conform to music theory. Finally, Costa et al. (2000) studied the effect of theoretical consonance on the expressive characteristics of the intervals. Musicians and nonmusicians were required to judge intervals on a semantic differential, made up of 30 bipolar adjective rating scales (e.g., pleasant/unpleasant). These scales were organized along three factors: emotional evaluation, activity, and potency. Results showed that, on scales testing the emotional factor, dissonant bichords were perceived as negatively polarized, while consonant bichords spread out over the neutral zone. The authors hypothesized that "emotional characterization of these (consonant) intervals is more influenced by the context in which they are arranged, than by the interval per se" (Costa et al., 2000). Interestingly, results did not show any influence of musical expertise.

While few experiments have been conducted to understand the perceptual, cognitive, and emotional aspects of consonance, even fewer studies have been aimed at analyzing its neurophysiological basis. Except for the positron-emission tomography (PET) experiment by Blood, Zatorre, Bermudez, and Evans (1999), which studied the cerebral blood flow changes related to affective responses to pieces of music that varied systematically in degree of consonance, the other studies have used the event-related brain potential (ERP) method. Because of its excellent temporal resolution, the ERP method provides a unique way to study perceptual, cognitive, and emotional processes as they unfold in time. It then becomes possible to precisely determine at which point in time the ERPs elicited in two experimental conditions start to diverge, and by inference, to determine when the two underlying processes start to differ. The experiments conducted so far with the ERP method mainly focused on consonance and dissonance of chords or notes presented in a musical context. Levett and Martin (1992) and Regnault, Bigand, and Besson (2001) showed that the occurrence of dissonant chords at the end of harmonic progressions is associated with a larger positive component, peaking around 600 ms (P600), compared to consonant chords. This P600 component has also been found in response to out-of-key notes (Besson & Macar, 1987; Besson & Faïta, 1995) or chords (Janata, 1995; Patel, 1998; but see also Koelsch, 2000, for other music-related electrophysiological components) within a musical context. Taken together, these results have led to the hypothesis that the P600 reflects the difficulty in integrating a note or a chord into a musical context, rather than the processing of dissonance per se. In order to exclude the potential influence of musical expectancy, it is therefore of interest to study the effect of consonance/dissonance out of a musical context.

The aim of the present experiment is twofold. First, in order to study the neurophysiological basis of consonance, it is important to determine whether consonant and dissonant chords are associated with different electrophysiological components or modulation of a same component when presented out of a musical context. To this end, we recorded the ERPs elicited by the different intervals of the chromatic scale. Participants had to judge, on a six-point scale, whether the intervals evoked a pleasant feeling or not. Based on intuitive knowledge and on the results of Costa et al. (2000), we expected dissonant intervals to be perceived as less pleasant than consonant ones. Of interest was to determine whether such a differential perception would be associated with different ERP effects. Second, it was of interest to test the hypothesis that, as proposed by Helmholtz (1877), the perception of dissonance results from the superposition of the partials of close frequencies. In order to do so, two notes, with different intervals of the chromatic scale, were played either together (harmonic intervals) or successively as melodic intervals. Since in this latter case the two notes are played at different points in time, the perception of roughness, if any, should be weaker than for harmonic intervals. Thus, the ERP effects associated with the sensory component of dissonance, if any, should be smaller for melodic than for harmonic intervals. Moreover, the differences between intervals should occur earlier, at a sensory stage of processing, for harmonic intervals than for melodic intervals. To be able to compare results for melodic and harmonic intervals, we classified melodic intervals according to the same rules as those typically used in music theory to classify harmonic intervals. This choice is supported by the theoretical work of Leman (1995), showing that the

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persistence of pitch in the short-term auditory store may give rise to consonance and dissonance sensations for melodic intervals, and by the experimental findings of Maher (1980) and Maher and Berlyne (1983), who showed that participants similarly categorized melodic and harmonic intervals. However, as their stimuli were pure tones, the possibility remains that the absence of harmonics might have artificially decreased the differences between the two types of intervals.

As mentioned above, the perception of consonance can result both from the physical characteristics of the sounds, played simultaneously or successively, and from the subjective "interpretation" of the sounds, based on our implicit or explicit knowledge of the musical system that we are most familiar with. In order to test these two possibilities, two types of analyses were computed on the ERP data. First, ERPs elicited by the different musical intervals were sorted as a function of their physical characteristics that, according to some music theories, distinguish three main categories of intervals: perfect consonance, imperfect consonance, and dissonance. Note, however, that results of psychoacoustic research have pointed out that the distinction between consonant and dissonant intervals is better considered as a continuum rather than as discrete categories. Thus, the classification of intervals that we used may seem arbitrary. However, it was useful for methodological reasons, linked with the need to average several trials from the same condition to obtain clear ERP data while maintaining the duration of the experiment at a reasonable length. Further, as mentioned above, it is in agreement with some music theories. Second, ERPs were sorted according to whether the intervals were judged as pleasant or unpleasant by the participants, independently of whether such categorization was in line with music theory or not. Hence, an effect observed only when data are sorted as a function of music theory (based on the acoustic properties of intervals) should reflect interval discrimination based on sensory processing, that is, on the interaction between the acoustic characteristics of the stimulus and the filtering properties of the auditory periphery. By contrast, an effect observed only when data are sorted as a function of the responses should reflect interval discrimination based on perceived consonance including both sensory and cognitive components. Taken together, results from both analyses should therefore allow us to disentangle bottom-up processes, linked with the basic acoustic characteristics of the intervals, from top-down processes responsible for the different feelings associated with the different intervals. Finally, as the perception of consonance may differ as a function of the expertise of the participants (Van de Geer, Levelt, & Plomp, 1962), we compared musicians, who had at least 10 years of music training, and nonmusicians, with no specific training and who were not music lovers.

Method

Participants

Twelve experienced musicians and 13 nonmusicians (mean age: 26 years old) participated in this study. However, 5 participants were excluded from further analyses due to a large number of artifacts (mainly blinks and eye movements), so that each group comprised 10 participants.

Stimuli

Stimuli consisted of series of two notes, composing the different intervals of the chromatic scale: unisons (1), minor and major seconds (2m and 2M), minor and major thirds (3m and 3M), fourth (4), fifth (5), minor and major sixths (6m and 6M), minor and major sevenths (7m and 7M), and octaves (8). As explained above, these intervals were grouped into three categories: Perfect Consonances (PC: 1, 4, 5, 8), Imperfect Consonances (IP: 3m, 3M, 6m, 6M), and Dissonances (D: 2m, 2M, 7m, 7M). In one part of the study, the notes were presented simultaneously (harmonic intervals), and in the other part, they were presented successively (melodic intervals). All participants listened to both harmonic and melodic intervals. The order of presentation of melodic and harmonic intervals was the same for all the participants, and the melodic intervals were presented first.[1]

Procedure

Intervals were presented in 13 blocks, corresponding to the 13 notes of the scale. Within each block, one note was chosen as the base note of the interval (it could be called the fundamental), and the 12 melodic intervals

[1]In a pilot experiment we found that when melodic intervals were presented after harmonic intervals, participants were likely to group the two successive notes that composed the melodic intervals together (i.e., imagining how they would sound if played together), as if they were harmonic intervals. Therefore, in order to avoid such a bias, we used a fixed melodic-harmonic order, with melodic intervals always played first. Although we do not believe that this is the case, we cannot dismiss the possibility that the order of presentation has produced some training bias.

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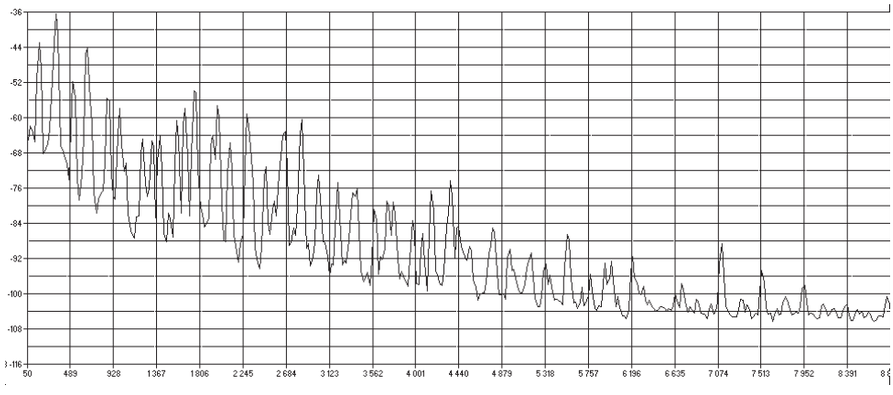

FIG. 1. Illustration of the frequency spectrum for a harmonic stimulus (F4 & E5).

(from the unison to the octave) or the 11 harmonic intervals (all but the unison) were built from this base note (e.g., minor second, fourth, etc.). Indeed, the perception of consonance of a given musical interval (A) may, to a large extent, depend on the interval that preceded it (B). In order to avoid such a context effect, it would be necessary to wait for the sensory trace associated with interval B to disappear. This, however, may take several seconds (Darwin, Turvey, & Crowder, 1972; Watkins & Watkins, 1980) and would make the whole study far too long and tiring. Thus, we decided to control and to minimize such context effects by presenting blocks of trials with a fixed base note. One potential problem with this solution is that the base note may induce a specific tonality within the block and intervals may possibly be judged according to this tonality. However, since the intervals were played in random order, the typical rules that characterize tonality (progressions, cadences) were absent and the context effects, if any, should be reduced to a minimum. In the melodic condition, the base note was always played first. Block order (change of the fundamentals) was also random.

Both the harmonic and melodic parts of the study were performed twice. In one case the base note was always lower than the other note of the interval, while in the other case it was always higher. This manipulation allowed using twice as many stimuli, thus increasing the ERP signal-to-noise ratio, while keeping note frequencies within two neighboring octaves. This last point is important since the sensation of roughness elicited by a given interval varies with frequency (Plomp & Levelt, 1965).

Thus, each participant was presented with $13 \times 12 \times 2$ melodic intervals (312) and $13 \times 11 \times 2$ harmonic intervals (286). Stimuli were generated using the piano sound produced by a general MIDI (Korg, X5DR) controlled by a 486 Compaq PC running under DOS (see Figure 1). Note frequencies ranged from 130.8 to 523.25 Hz and the equal-tempered system of tuning was used. Note durations were 350 ms and, for melodic intervals, they were separated by a silence of 750 ms. Participants were asked to judge whether the musical intervals were pleasant or not. They used a six-point scale (1 very pleasant; 6 very unpleasant) to give their answers. Although we are aware that collapsing reduces sensitivity, this was done for three main reasons. First, to have the same number of categories in music theory. Second to have enough trials in the ERP averages for each condition. Third, to simplify presentation of the results. The six response types were collapsed into three categories: Very Pleasant (VP: 1-2), Quite Pleasant (QP: 3-4), and Unpleasant (U: 5-6).

Data Acquisition and Analysis

EEG was recorded for 2600 ms (harmonic intervals) or 3700 ms (melodic intervals) starting 200 ms before the onset of the stimulus (see Figure 2), from 28 scalp electrodes, mounted on an elastic cap and located at standard left and right hemisphere positions over frontal, central, parietal, occipital, and temporal areas (International 10/20 system sites: Fz, Cz, Pz, Oz, Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fc5, Fc1, Fc2, Fc6, Cp5, Cp1, Cp2, Cp6).

These recording sites, plus an electrode placed on the right mastoid, were referenced to the left mastoid electrode; the data were then re-referenced offline to the algebraic average of the left and right mastoids. Impedances of these electrodes never exceeded 3 kΩ. In order to detect blinks and vertical eye movements, the

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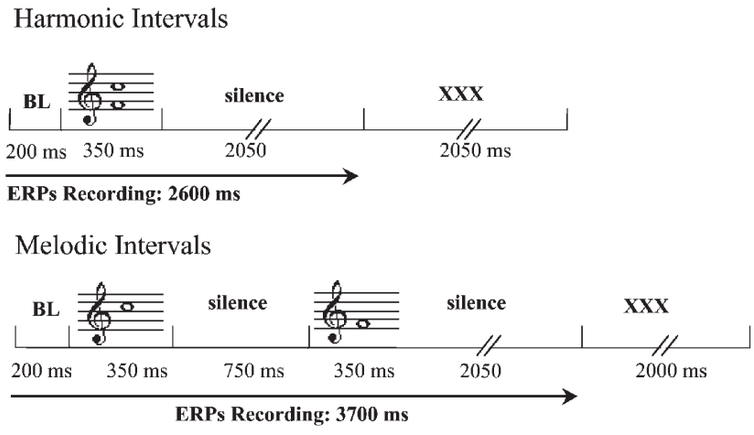

FIG. 2. Illustration of the sequence of events within a trial for both melodic and harmonic intervals (BL = baseline).

horizontal electrooculogram (EOG) was recorded from electrodes placed 1 cm to the left and right of the external canthi, and the vertical EOG was recorded from an electrode beneath the right eye, referenced to the left mastoid. Trials containing ocular artifacts, movement artifacts, or amplifier saturation were excluded from the averaged ERP waveforms. The EEG and EOG were amplified by a SA Instrumentation amplifier with a pass band of 0.01-30 Hz and were digitized at a sampling rate of 250 Hz by a PC-compatible microcomputer (Compaq Prosignia 486).

ERP data were analyzed by computing the mean amplitudes (i.e., integration of amplitude values over time) in successive 100 ms latency bands from stimulus onset and in specific latency ranges selected according to visual inspection of the ERP traces. The mean amplitudes were submitted to a four-way mixed analysis of variance (ANOVA), including musical expertise (musicians vs. nonmusicians) as a between-subjects factor, and base note (high vs. low), degree of consonance (PC vs. IC vs. D) or response (VP vs. QP vs. U), and electrodes (26 levels[2]) as within-subjects factors. The associated p-values were adjusted with the Greenhouse-Geisser epsilon correction for nonsphericity (ε). Reported are the uncorrected degrees of freedom and the probability level after correction. Tukey tests were used to test the significance of post hoc comparisons (the letter q indicates the value for this test).

Behavioral Results

Behavioral data are presented for the three categories of stimuli (Perfect Consonances, Imperfect Consonances, and Dissonances) and for the three types of responses given by participants (Very Pleasant, Quite Pleasant, and Unpleasant). Since no significant difference was found as a function of whether the base note was higher or lower than the changing note, $\chi^2(2) = 3.66$, $p > 0.1$, results from these two subcategories were averaged together. Percentages of responses are reported in Table 1.

For musicians, the distribution of the responses within each category—Perfect Consonance (PC), Imperfect Consonance (IC), and Dissonance (D)—is in line with music theory (i.e., the values of the diagonal are the highest in Table 1). Thus, for both melodic and harmonic intervals, PC intervals are most often categorized as Very Pleasant, IC as Quite Pleasant, and D as Unpleasant (comparison of the distribution of the responses for the expected category versus the non-expected categories always gives $\chi^2(2) > 60$, $p < .001$).[3] However, nonmusicians have a tendency to use the central points of a scale more than musicians, so that the different intervals—PC, IC, and D—are most often categorized as Quite Pleasant. Indeed, within categories, the judgment of nonmusicians does not follow the classification of intervals according to music theory

[2]While EEG was recorded from 28 electrodes, 26 electrodes were included in the ANOVA because the ERP traces from Fp1 and Fp2 showed a large positive shift that was not representative of recordings at other scalp sites.

[3]Note that the terms "expected" and "unexpected" categories are to be considered in line with music theory so that PC intervals are expected to be categorized as very pleasant, IC as quite pleasant, and D as unpleasant.

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TABLE 1. Correspondence between music theoretic and response categories. Percentages and number of responses (in brackets) within each category/response type.

	Musicians			Nonmusicians		
Category	Very Pleasant	Quite Pleasant	Unpleasant	Very Pleasant	Quite Pleasant	Unpleasant
Melodic intervals						
Perfect consonance	67 (627)	26 (243)	6 (56)	35 (328)	43 (402)	22 (206)
Imperfect consonance	26 (243)	53 (496)	21 (197)	25 (234)	48 (449)	27 (253)
Dissonance	17 (159)	39 (365)	44 (412)	24 (225)	43 (402)	33 (309)
Harmonic intervals						
Perfect consonance	54 (463)	38 (326)	9 (76)	38 (326)	46 (395)	16 (137)
Imperfect consonance	31 (266)	57 (489)	12 (103)	27 (232)	50 (429)	22 (189)
Dissonance	9 (77)	31 (266)	59 (506)	13 (112)	41 (352)	46 (395)

Note. PC = perfect consonance; IC = imperfect consonance; D = dissonance; VP = very pleasant; QP = quite pleasant; U = unpleasant.

(comparison of the distribution of the responses for the expected category versus the nonexpected categories always gives $\chi^2(2) > 25$, $p < .001$), except for the harmonic dissonant category (not significant, $p > .1$). However, looking at the columns, it is clear that the percentage of responses for each response type is always highest in the expected category (comparison of the distribution of the expected responses across categories versus the nonexpected response types always gives $\chi^2(2) > 10$, $p < .01$, except for the Quite Pleasant response with harmonic intervals $p > .1$).

Discussion

Musicians tended to categorize harmonic and melodic intervals according to music theory. Nevertheless, 44% of the intervals were not categorized according to this criterion. Moreover, nonmusicians did not follow the rules of tonal music and showed a strong tendency to categorize most of the intervals in the intermediate category, that is, as quite pleasant. These results thus point to the importance, for both musicians and nonmusicians, of averaging ERPs according to not only music theory but also participants' responses.

Comparison of the results for the melodic and harmonic intervals showed that, for musicians, perfect consonances were more often judged as very pleasant for melodic (67%) than for harmonic intervals (54%, $\chi^2(1) = 24.7$, $p < .001$). Note, however, that this is mostly due to the fact that while perfect consonances in melodic intervals also included the unison, the most consonant interval, this was not the case for harmonic perfect consonant intervals. Indeed, when the unison is removed from the analyses, the percentage of melodic intervals judged very pleasant drops to 61% (however, the difference remains significant, $\chi^2(1) = 9.2, p < .005$).

Dissonant intervals were more often judged as dissonant for harmonic than for melodic intervals, and this was found for both musicians (59% vs. 44%, $\chi^2(1) = 9.6, p < .005$) and nonmusicians (46% vs. 33%, $\chi^2(1) = 10.5$, $p < .001$). This result conforms to our prediction, based on the theory of acoustic roughness, following which the beating produced by the simultaneous emission of two sounds with close partials will produce a sensation of roughness, unpleasant to the ear. However, roughness alone cannot explain why the melodic dissonant intervals were also judged as unpleasant, since the notes were separated by a 750 ms silence. It may be that musicians transpose their harmonic knowledge to melodic intervals, so that they apply the same rules in both cases. Whereas nonmusicians have no explicit harmonic knowledge that would help them correctly recognize and assign the intervals to the appropriate theoretical category, an interval-by-interval analysis (see Figure 3) suggests that nonmusicians did use a different strategy. They seem to rely on a principle of proximity: the farther away, the more unpleasant. Thus, for instance, while musicians considered 80% of the melodic octave intervals as pleasant, nonmusicians considered only 20% of the melodic octave intervals as pleasant, $\chi^2(1) = 46.8$, $p < .001$, whereas they judged the harmonic octave as pleasant on 50% of the responses (see Figure 3). Finally, in contrast to melodic intervals, the pattern of responses for the harmonic intervals was similar for both musicians and nonmusicians.

Electrophysiological Results

Results for both harmonic and melodic intervals are reported in turn, with a short discussion for each type of interval. Statistical analyses were performed both as a

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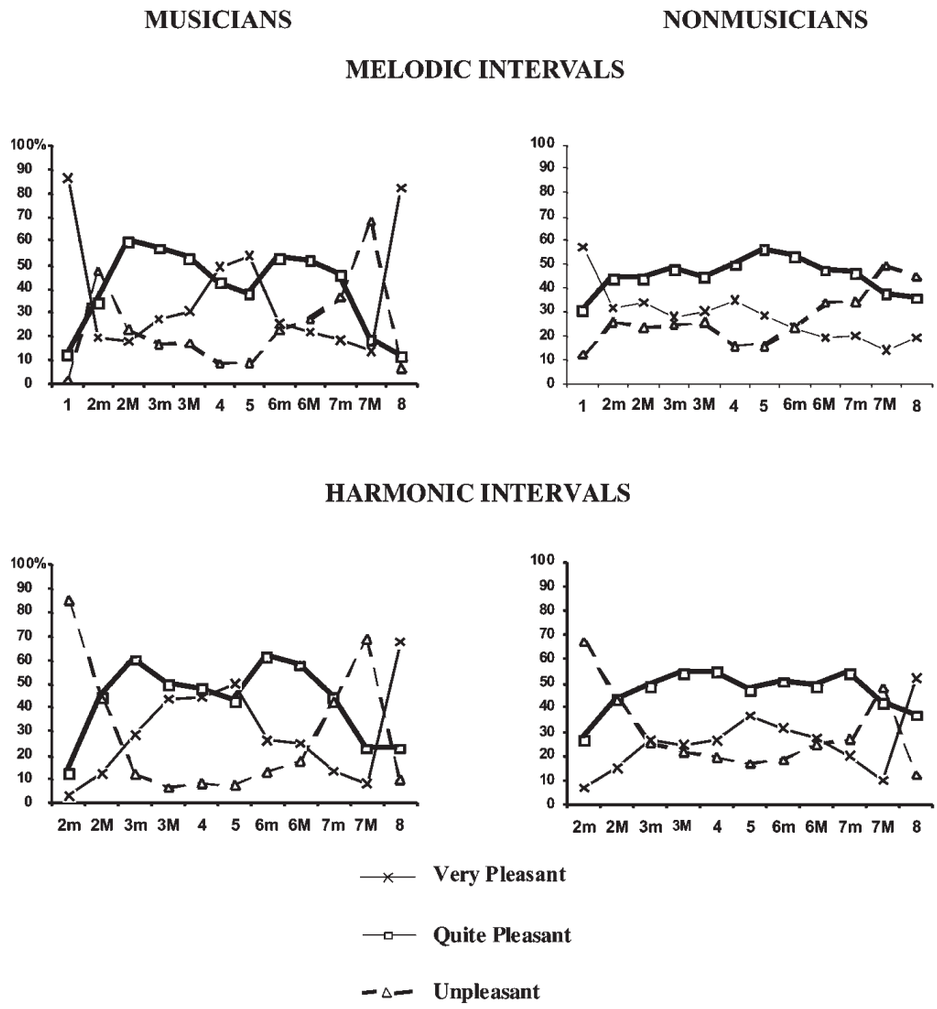

FIG. 3. Percentages of responses within each category (very pleasant, quite pleasant, unpleasant) and for each type of musical interval (1: unison; 2m: minor second; 2M: major second; 3m: minor third; 3M: major third; 4: fourth; 5: fifth; 6m: minor sixth; 6M: major sixth; 7m: minor seventh; 7M: major seventh; 8: octave).

function of music theory and as a function of the responses given by the participants. In both cases, each of the ERP traces comprises a large number of trials (mean = 824; s.d. = 156). Results of the ANOVAs computed in different latency bands are only reported when they are significant.

Harmonic Intervals

AVERAGED ACCORDING TO MUSIC THEORY

As can be seen from Figure 4, the ERPs elicited in the three categories perfectly overlap until 200 ms post-stimulus onset. Accordingly, results of the ANOVAs revealed no significant main effect or interaction in the 0-200 ms range. The ERPs then start to diverge, particularly for the nonmusicians, resulting in a significant Expertise × Consonance interaction, in the 200-300 ms range, $F(2, 36) = 3.81$, $\varepsilon = 0.92$, $p < .05$. Results of post hoc Tukey tests revealed that for nonmusicians the N2 component was significantly larger for perfect consonances (0.86 μV) than for both dissonances (1.66 μV; $q(2, 18) = 3.49$; $p < .05$) and imperfect consonances (1.92 μV; $q(2, 18) = 4.69$; $p < .01$). No significant differences were found for musicians in this latency band. In the 400-600 ms latency band, the consonance by electrode interaction was significant,

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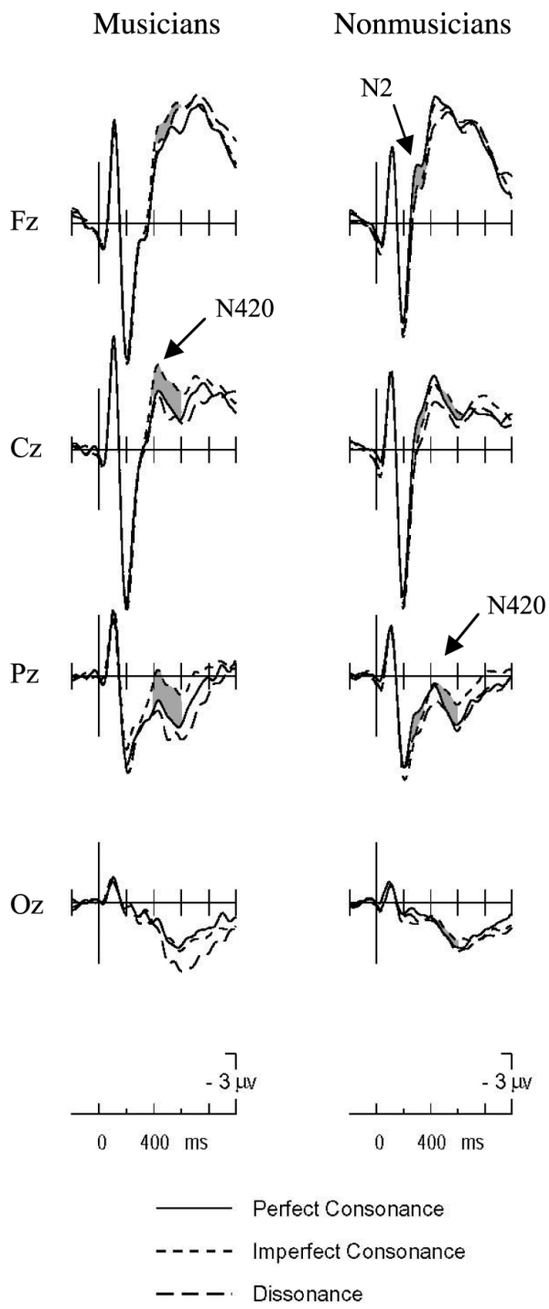

FIG. 4. ERPs elicited by the presentation of harmonic intervals to musicians and nonmusicians at four midline sites (Fz, Cz, Pz, and Oz). ERPs are averaged according to music theory that defines three main categories of intervals: perfect consonance, imperfect consonance, and dissonance. On this figure, as on the following ones, the time (in ms) from stimulus onset is represented on the abscissa, and the amplitude of the effect (in µV) is on the ordinate. Negativity is up. The shaded areas indicate the latency bands in which the ERP differences were found to be significant.

$F(50, 900) = 2.05$, $\varepsilon = 0.19$, $p < .05$. Over centro-parietal regions, imperfect consonances were associated with a larger negativity, peaking around 420 ms (−1.39 µV), than were perfect consonances (−0.87 µV) and dissonances (−0.75 µV).

AVERAGED ACCORDING TO PARTICIPANTS' RESPONSES

As can be seen from Figure 5, ERPs are similar to those obtained in the averages according to music theory, except for one interesting difference. The amplitude of the N100 component differs as a function of the responses, so that the main effect of response type was significant in the 100-200 ms latency band, $F(2, 36) = 3.9$, $\varepsilon = 0.77$, $p < .05$: overall, the N1-P2 complex was more positive for the intervals judged as very pleasant (1.36 µV) than for those judged as unpleasant (0.35 µV; $q(2, 36) = 3.78$; $p < .05$). Separate ANOVAs showed that this effect was significant only for musicians, $F(2, 18) = 3.73$, $\varepsilon = 0.99$, $p < .05$; nonmusicians: $F(2, 18) = 1.40$, $\varepsilon = 0.95$, $p > .20$. As found for the ERPs averaged according to music theory, the main effect of response type was also significant in the 400-600 ms latency band, $F(2, 36) = 3.84$, $\epsilon = 0.93$, $p < .05$: intervals judged as quite pleasant elicited a larger negative component (−1.56 µV) than intervals judged as very pleasant (−0.43 µV; $q(2, 36) = 3.51$; $p < .05$) or unpleasant (−0.95 µV; though here $p > .05$).

Discussion

When ERPs were averaged according to music theory, the N200 component elicited by nonmusicians was largest for the perfect consonances, that is, for the intervals that, according to the theory of consonance developed by Helmholtz (1877), present a minimal roughness. The occurrence of an N2 component has often been associated with sensory discrimination and categorization processes (Ritter, Simson, & Vaughan, 1983; Ritter, Simson, Vaughan, & Friedman, 1979; Simson, Vaughan, & Ritter, 1977). Thus, in our experiment, the largest N200 to perfect consonance may reflect the categorization of the intervals on the basis of their acoustic characteristics (roughness). However, this interpretation seems in contradiction with the behavioral results showing that, overall, nonmusicians did not categorize intervals according to music theory. It may be that nonmusicians first perceive the differences in the acoustical properties of the sounds but that they do not rely on the result of this perceptual analysis because they lack confidence in their perceptual judgment. As mentioned above, they rather seem to adopt the strategy of categorizing intervals in the

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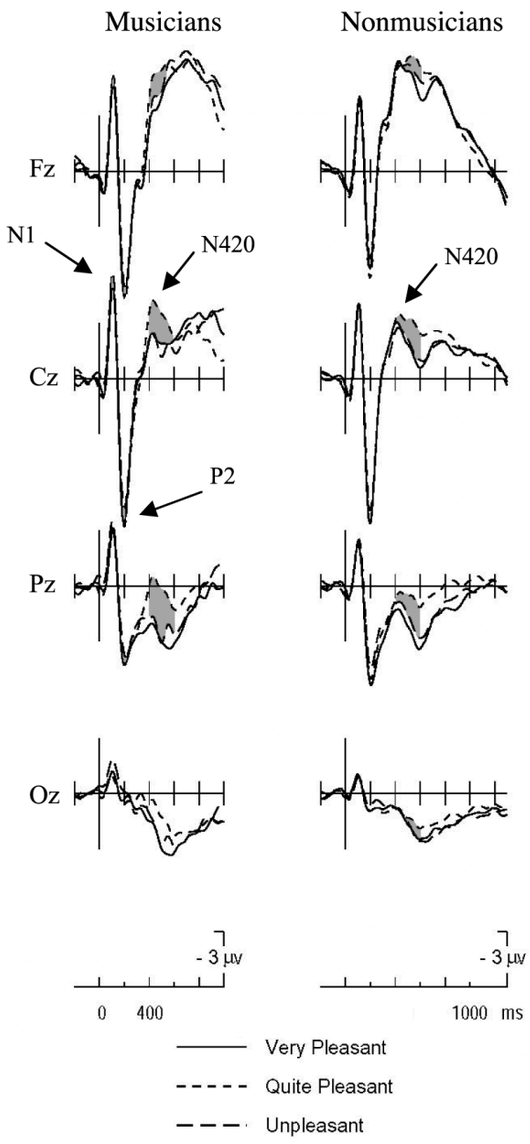

FIG. 5. ERPs elicited by the presentation of harmonic intervals to musicians and nonmusicians at four midline sites (Fz, Cz, Pz, and Oz). ERPs are averaged as a function of the responses given by the participants to categorize the musical intervals as very pleasant, quite pleasant, and unpleasant.

intermediate category. In contrast to nonmusicians, musicians showed no difference in the early latency bands (0-300 ms).

Interestingly, results are quite different when averaged according to the responses. In this case, musicians showed an early effect of consonance on the N1-P2 complex, that is, between 100 and 200 ms post-stimulus onset. As this effect was not observed when ERPs were averaged according to music theory, it is more likely to reflect the judgment of the intervals on the basis of their cognitive/affective characteristics than on the basis of their acoustical properties. Thus, musicians seem able to quickly categorize harmonic intervals as a function of the feeling, pleasant or unpleasant, that they evoke. Moreover, they may have built expectations of what they consider pleasant or unpleasant, and this may consequently influence perceptual processing at the level of the auditory cortex. In line with this hypothesis, several studies have shown that the organization of the auditory cortex can be modified by learning (i.e., long-term plasticity; see Rauschecker, 1999; Pantev & Lütkenhöner, 2000).

Taken together, our results indicate that the ERPs elicited by the different musical intervals diverge earlier when the judgment is made on the basis of subjective consonance (100-200 ms) than when the judgment results from the analysis of their acoustic characteristics (200-300 ms). Note, however, that the results reported here for musicians differ in some respects from the results of a previous experiment in which we examined the effects of sensory consonance and harmonic context (Regnault, Bigand, & Besson, 2001). In this previous experiment, we found that the N1 component was larger in response to consonant than to dissonant chords and that the P2 component (100-200 ms latency range) was larger in response to dissonant than to consonant chords. By contrast, the present results revealed no significant difference in the averages as a function of music theory, which is the relevant comparison with the Regnault et al. results (2001). Given that the main difference between experiments is that chords are presented within a musical context in the Regnault et al. study, and out of musical context in the present study, it may be that this factor accounts for the different results reported in both experiments.

Note, moreover, that results also showed an interaction between the effects of consonance and musical expertise, so that music training was associated with qualitative changes in the processing of harmonic intervals. Indeed, musicians discriminated intervals at an early stage and on the basis of perceived consonance, as revealed by significant differences between intervals judged as pleasant or unpleasant in the 100-200 ms latency band (see Figure 5). By contrast, nonmusicians discriminated the intervals later, on the basis of their acoustic characteristics, as revealed by significantly larger N2 components for perfect than for imperfect consonances and dissonances in the 200-300 ms latency band (see Figure 4).

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Finally, and perhaps most importantly, a consonance (or response type) effect was also found in the 400-600 ms latency band, both in the ERP averaged as a function of music theory and in the ERP averaged as a function of the responses. The relatively late latency of this negativity, peaking around 420 ms post-stimulus onset, indicates that it is more likely to reflect cognitive than perceptual processing. The amplitude of this negative component was largest for the imperfect consonances, which are the most difficult intervals to categorize. Thus, the possibility that the negativity reported in this experiment is functionally similar to the well-known N400 reported in language experiments (Kutas & Hillyard, 1980) should be examined. The finding, reported in many experiments, that the amplitude of the N400 varies as a function of the difficulty of the integration process has led us to consider the N400 component as a good index of the integration of a word meaning when the word is presented in isolation, or of the integration of meaning within a sentence context, when words are presented in the context of other words (see Besson, Magne, & Regnault, in press, for a review). However, two main arguments lead us to consider that the negative component reported here is functionally different from the N400. First, no clear meaning is associated with the musical intervals used in the present experiment; second, these intervals need not be integrated within a musical context. Thus, rather than reflecting the difficulty of the integration process, the negative component reported here is more likely to reflect the difficulty of the categorization process. In this respect, it may be functionally similar to the N200 component, delayed in latency, because intervals in the intermediate category were more difficult to categorize than intervals belonging to the other two categories.

Melodic Intervals

AVERAGED ACCORDING TO MUSIC THEORY

No significant main effect or interactions were found over the entire ERP recording epoch.

AVERAGED ACCORDING TO PARTICIPANTS' RESPONSES

As can be seen in Figure 6, no significant main effect or interaction was found prior to the second note and in the 400 ms following its presentation. However, in the 400-500 ms latency range following the second note, results showed a significant main effect of response type, $F(2, 36) = 3.56$, $\varepsilon = 0.92$, $p < .05$. Intervals judged as quite pleasant were associated with a larger negativity (0.33 μV) than those judged as very pleasant (1.44 μV; $q(2, 36) = 3.57$; $p < .05$). The intervals judged as unpleasant were associated with an intermediate negativity (1.21 μV) that did not differ statistically from that of the other two conditions.

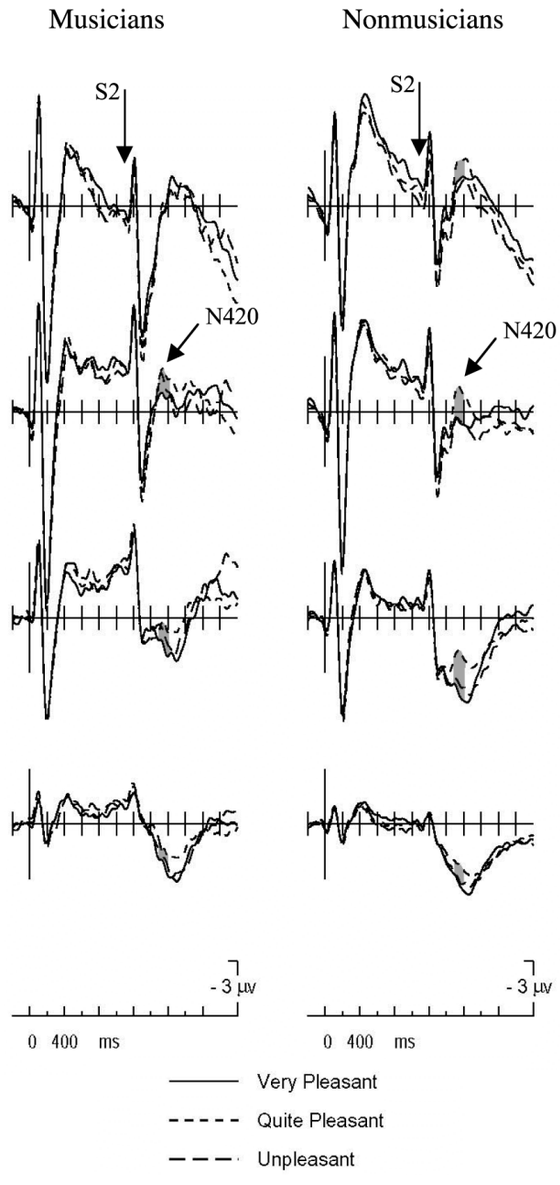

FIG. 6. ERPs elicited by the presentation of melodic intervals to musicians and nonmusicians at four midline sites (Fz, Cz, Pz, and Oz). ERPs are averaged according to music theory. The arrow (S2) points to the moment in time when the second note of the melodic interval is presented.

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Discussion

When data are averaged as a function of music theory, results do not show any significant effect of consonance. Thus, two notes separated by a silence seem to be processed similarly whatever the interval between them. By contrast, when data are averaged as a function of participants' responses, the effect of response type is significant. However, the relatively late latency (400-600 ms) of this effect suggests a cognitive rather than a perceptual process. As mentioned above, the decision to be made was probably more difficult for the intervals in the intermediate category than for the intervals in the other two categories. This is not surprising insofar as the characteristics of perfect consonances and dissonances are more salient. Thus, our results seem to indicate that, in line with our hypotheses, perceived consonance of melodic intervals did not influence sensory/perceptual processing of notes but rather later cognitive computations. Finally, it is interesting to note that the effect of musical expertise was not significant in any latency band considered for analysis.

General Discussion and Conclusion

One major aim of our experiment was to investigate whether the processing of consonance/dissonance would be associated with specific ERP effects. Our results show differences depending on the consonance of musical intervals, for both musicians and nonmusicians. Moreover, the effect of consonance on the processing of musical intervals differs according to expertise and to the mode of interval presentation (i.e., melodic vs. harmonic intervals). Finally, the comparison of our results with those of previous experiments clearly points to the strong influence of musical context. These results add arguments in favor of both a physiological origin and a psychological origin (linked with the listener's expertise and cultural factors) to the sensation of consonance. It is, however, difficult to propose one single model that could account for the different aspects involved in the processing of consonance. Indeed, several factors, such as musical expertise, the simultaneous or successive presentation of notes, and the musical context, lead to qualitatively different neural processing. Thus, even one of the most basic perceptual characteristics of sounds, consonance, does not involve a simple process. Our results point to the complexity of the cerebral processes involved in sound perception and, even more so, in music perception. In the following we will consider the effect of consonance on the processing of intervals as a function of three factors: expertise, simultaneity, and context.

Expertise

Our results clearly show an influence of musical expertise on the perception of consonance when intervals are presented out of a musical context. While nonmusicians process harmonic intervals differently as a function of their acoustic characteristics, musicians discriminate musical stimuli earlier as a function of perceived consonance. These two types of categorization are reflected by different variations of the electrophysiological components elicited by perfect consonances: a larger N2 component for nonmusicians and a more positive N1-P2 complex for musicians. Auditory cortex plasticity may, at least to some extent, explain this effect. Functional reorganization of the auditory cortex after cochlear lesions or repetitive auditory stimulation has been shown in animals (see Rauschecker, 1999, for a review). Moreover, Weinberger (1998) was able to produce conditioning in guinea pigs by associating the presentation of a sound of given frequency with an aversive stimulus. Results showed an increase of the size of the receptor fields in the primary auditory cortex that corresponded to the conditioned sounds, while the size of receptor fields corresponding to nonconditioned sounds decreased. The author interpreted these data in terms of "physiological memory of auditory cortex," reflecting the "acquired meaning of a stimulus." Thus, some type of functional reorganization of the auditory cortex, as found by Weinberger (1998) in animals, may occur in humans. Music training, especially when started at an early age, by favoring a long and sustained associative learning between a given interval and an emotion, may increase the cortical representations of sounds for musicians. Such results have indeed been reported in the literature (see for instance Schlaug, Jancke, Huang, Staiger, & Steinmetz, 1995; Pantev & Lütkenhöner, 1998; Pantev, Roberts, Schulz, Engelien, & Ross, 2001).

Simultaneity

Results differed for harmonic and melodic intervals. Whereas for harmonic intervals the consonance effects start as early as 100 ms (for musicians in the averages as a function of the responses) or 200 ms (for nonmusicians in the averages as a function of music theory) after stimulus onset, no effect is found before 400 ms for melodic intervals. Thus, as expected on the basis of the theory of consonance, the sensation of roughness seems intrinsically linked with the simultaneous presentation of notes, out of musical context. In other words, the effect of consonance, whether reflecting

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stimulus categorization based on learning (musicians) or on acoustical characteristics (nonmusicians), seems to mainly rely on the simultaneity of partials of neighboring frequencies. Consequently, these results demonstrate that the physical characteristics of sounds are at least partly responsible for the sensation of consonance/dissonance. Thus, these results are in line with Helmholtz's theory of consonance. However, analyses of behavioral responses show that even when notes are presented in succession (i.e., melodic intervals), musicians are able to categorize them according to music theory. Taken together, ERPs and behavioral results underlie the implication of both physical and psychological factors in the sensation of consonance.

Context

Our results show that the consonance of intervals presented out of musical context modulates the amplitude of an N1/P2 complex, an N2 component, and a negative component peaking between 400 and 600 ms post-stimulus onset. These negative components differ from the effect of consonance found when chords are presented in a musical context. Indeed, as mentioned in the introduction, dissonant chords ending musical phrases elicit the occurrence of positive components, with different latencies but often referred to as P600 components (Levett & Martin, 1992; Regnault et al., 2001). Note, however, that a positive component, peaking around 600 ms post-stimulus onset, can be seen in response to both harmonic and melodic intervals in both musicians and nonmusicians. Interestingly, the amplitude of this positivity is larger in response to perfect consonance and dissonance than to imperfect consonances when ERPs are averaged as a function of music theory (see Figure 4). Similarly, its amplitude is larger for very pleasant and unpleasant intervals when ERPs are averaged as a function of the participants' responses (see Figures 5 and 6). Therefore, such an ordering does not correspond to what would be expected based on the literature related to the musical P600. Rather, one would expect the dissonant (or unpleasant) intervals to be associated with the largest positivity, the imperfect consonance (or quite pleasant) intervals to be associated with intermediate positivity, and the perfect consonance (or very pleasant) intervals to be associated with the smallest positivity. Because we did not find such relationships, we tend to consider that the interpretation proposed above, in terms of categorization difficulty of imperfect consonance (or quite pleasant) intervals compared to the other two categories, accounts better for our pattern of results. Taken together, the findings reported in the literature and the present results underline the importance of a musical context in generating specific expectancies regarding the acoustical characteristics of chords. When no musical context is presented, as in this experiment, the functional interpretation of the late positivity may differ from the current functional interpretation of the P600 in terms of the integration cost of an unexpected event in a musical context (Patel, 1998).

Author Note

Both Daniele Schön and Pascaline Regnault equally contributed to this work.

This work was completed while Daniele Schön was a postdoctoral student, first supported by a grant of the International Foundation for Music Research (IFMR) to Mireille Besson (RPA #194) and later by a grant from the Human Frontier Science Program (HFSP, RG P0053/2002). We thank Stephen McAdams, Action Editor, and three anonymous reviewers for helpful comments on a previous version of the manuscript.

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